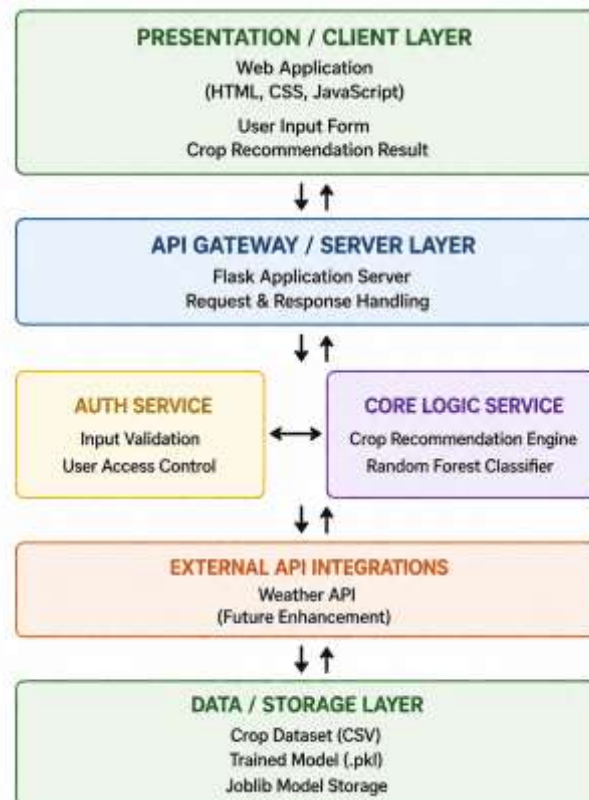


Solution Architecture

Date	15 March 2026
Team ID	XXXXXX
Project Name	OPTICROP – SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides user interface for entering soil and weather parameters and displaying crop recommendations	HTML, CSS, JavaScript
API Gateway	Receives requests from users and communicates with the machine learning model	Flask
Microservices / Core Logic Service	Processes input data and generates crop recommendations using machine learning algorithms	Python, Scikit-Learn, Random Forest
Database / Data Layer	Stores agricultural dataset and trained machine learning model files	CSV Dataset, Joblib